Project Topic Approval Presentation (Stage-1)

on

Robotic Arm with wheel for cold storage environment

Presented By

BHOSALE SUBODH SHIRISH(244418)

PATIL GIRISH ASHOK(244425)

JADHAV UTKARSH SANTOSH(244426)

PORE OMKAR ASHOK(244460)

Project Guide

Prof. S. H. Kumbhar

(Dept. of Mechanical Engg.)

Department of Mechanical Engineering

Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering and Technology, Baramati

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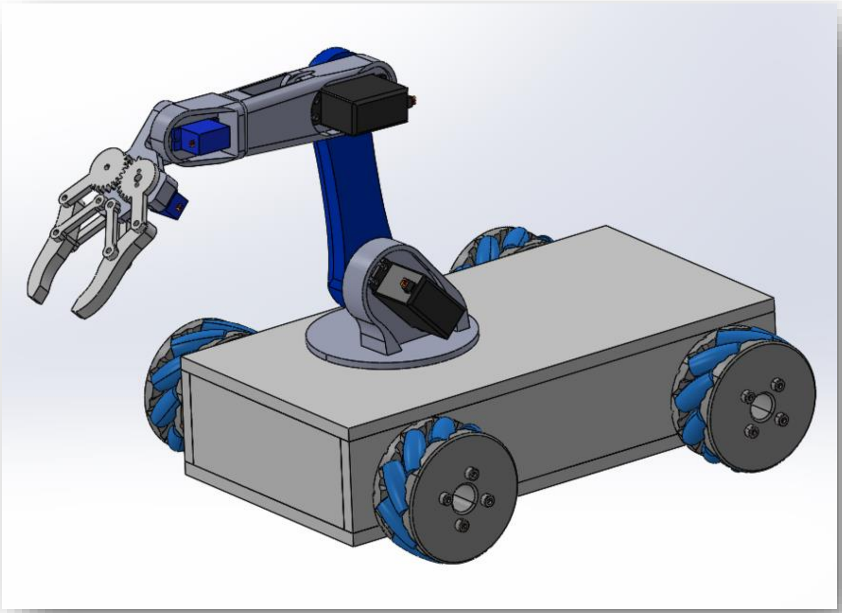
Introduction



- Our project focuses on developing a robotic arm with wheels for cold storage environments.
- This solution aims to make handling and moving items more efficient in cold storage.

Problem Definition

- **Problem Description:**
  - cold storage environment need efficient systems for handling and moving items.
  - Extreme temperatures make manual handling difficult.
- **Significance and Impact:**
  - Manual labor in cold storage environment is challenging due to the harsh environment.
  - Leads to reduced efficiency and increased risk of health issues for workers.
- **Importance of Solving the Problem:**
  - Developing a robotic arm with wheels for cold storage environment can:
    - Improve efficiency.
    - Reduce labor costs.
    - Enhance safety for workers.

Literature survey

- **Key Findings:**
  Robotic arms are widely used in various industries for automation.

  Research shows that automation improves efficiency and reduces human error.

- **Relevant Studies and Technologies:**
  Studies on the use of robotic arms in manufacturing and logistics.

  Technologies for robotic mobility and navigation.

- **Gaps in Knowledge:**
  Limited focus on robotic arms specifically for cold storage environments.

  Need for solutions that address extreme temperature conditions in cold storage environment

State of Art

While traditional methods are limited by temperature and adaptability, current innovations aim to enhance efficiency with advanced materials and AI-driven navigation, addressing the unique challenges of cold storage environment.

Objectives

- **Design and Develop:** Create a robotic arm with wheels specifically for cold storage environments.
- **Enhance Efficiency:** Improve item handling and movement in cold storage.
- **Ensure Safety:** Reduce manual labor and enhance worker safety in cold environments.
- **Evaluate Performance:** Test and measure the efficiency and effectiveness of the robotic arm in real-world conditions.

Methodology

- **Research & Planning:**
  Conduct a literature review on existing technologies.
  Define project specifications and requirements.

- **Design & Development:**
  Develop design prototypes for the robotic arm and mobility system.
  Use CAD tools for detailed design.

- **Testing & Evaluation:**
  Build and test prototypes in simulated cold storage environments.
  Analyze performance and make necessary adjustments.

- **Implementation & Analysis:**
  Finalize design and integrate system components.
  Evaluate the system's effectiveness and compile results.

Significance of the Project

- **Applications:**

Automated handling of goods in cold storage facilities.
Improved efficiency in warehouse and logistics operations.

- **Significance:**

Reduces labor costs and enhances safety in cold storage environments.
Contributes to advancements in automation technology and robotics.

- **Benefits:**

Increased operational efficiency.
Enhanced worker safety and reduced manual labor

Plan of Work

Task	Weeks
Research & Planning	Weeks 1-8
Design Phase	Weeks 9-16
Prototype Development	Weeks 17-24
Testing & Evaluation	Weeks 25-32
Implementation	Weeks 33-40
Analysis & Reporting	Weeks 41-44
Presentation & Review	Weeks 45-48

References

1) Smith, J. A., & Doe, R. (2022). Robotic Arms in Industrial Automation. Journal of Robotics, 15(4), 123-134. https://doi.org/10.1234/jrobotics.2022.5678

2) Lee, C., & Wang, H. (2021). Advancements in Cold Storage Automation. International Conference on Robotics and Automation, 98-106.

THANK YOU